Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 6

Biconditionals, and a Textbook Claim Worth Checking

Covers Chartrand & Zhang, Section 1.4

Status of this set

Ungraded practice. Solutions released with it. Attempt everything before reading any solution. Problem 8 is unusual — read its instructions carefully.

Part I: How to Practice This Material

A biconditional is two claims wearing one coat

The only two facts you need

1. $P \Leftrightarrow Q$ is true exactly when P and Q have the **same truth value**. To evaluate one, find both truth values and compare. Nothing else is required.
2. $P \Leftrightarrow Q$ is $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$. To *prove* one, you must prove **both** implications. To *refute* one, a single failing direction is enough.

Three mistakes to expect

1. **Thinking a true biconditional needs both sides true.** Two false statements also agree, so the biconditional holds.
2. **Proving one direction and stopping.** Half a proof of an "iff" is not a proof.
3. **Assuming a printed claim is correct.** Problem 8 exists because of this one.

Part II: Practice Problems

Section A — Evaluating biconditionals

1. Determine whether each is true or false, with a one-line reason.

(a) 11 is prime if and only if 12 is composite.

(b) $2^4 = 16$ if and only if $4^2 = 8$.

(c) $\sqrt{2}$ is rational if and only if 0 is negative.

(d) 6 is even if and only if 6 is prime.

2. For an integer n, let $P(n)$: n is divisible by 6, and $Q(n)$: n is divisible by both 2 and 3.

(a) Evaluate $P(n) \Leftrightarrow Q(n)$ for $n = 12$, $n = 9$, and $n = 5$.

(b) Do you believe $P(n) \Leftrightarrow Q(n)$ holds for every integer n? Say what you would have to prove to establish it.

Section B — Saying it in words

3. For an integer n, let $P(n)$: $5n + 3$ is even, and $Q(n)$: n is odd. State $P(n) \Leftrightarrow Q(n)$ (a) using "if and only if" and (b) using "necessary and sufficient."

4. Rewrite each as a biconditional using "if and only if."

(a) A valid session token is necessary and sufficient for access to be granted.

(b) If the queue is empty the worker sleeps, and if the worker sleeps the queue is empty.

Section C — Connections back to Chapter 1

5. Use a truth table to verify that $P \Leftrightarrow Q \equiv (P \wedge Q) \vee \big((\sim P) \wedge (\sim Q)\big)$.

P	Q	$P \Leftrightarrow Q$	$P \wedge Q$	$(\sim P) \wedge (\sim Q)$	$(P \wedge Q) \vee \big((\sim P) \wedge (\sim Q)\big)$
T	T				
T	F				
F	T				
F	F				

6. Explain, in two or three sentences, why $\sim(P \Leftrightarrow Q) \equiv P \oplus Q$ without building a new truth table.

Section D — Writing

7. In five to eight complete sentences, explain why proving a biconditional requires two separate arguments, while proving an implication requires only one. Illustrate with the claim "an integer n is odd if and only if n^2 is odd," and say specifically what the two arguments would have to establish.

Section E — Challenge (optional, and unusual)

8. Exercise 24(e) in Section 1.3 of your textbook instructs you to *verify* the following by truth table:

$$(P \Rightarrow Q) \wedge (Q \Rightarrow R) \;\equiv\; P \Rightarrow R.$$

Build the table. Report exactly what you find, and do not adjust your answer to match what the book expects.